

Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;
import java.io.*;

public class Main {
    static ArrayList<String> idList = new ArrayList<>();
    static HashMap<String, String[]> patientMap = new HashMap<>();
    static StringBuilder out = new StringBuilder();

    public static void main(String[] args) throws IOException {
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        int choice;
        do {
            choice = Integer.parseInt(br.readLine().trim());
            switch (choice) {
                case 1: registerPatient(br); break;
                case 2: searchPatient(br); break;
                case 3: displayAll(); break;
                case 4: deptCount(); break;
                case 5: break;
            }
        } while (choice != 5);

        System.out.print(out.toString().stripTrailing());
    }

    static void registerPatient(BufferedReader br) throws IOException {
        String id = br.readLine().trim();
        String name = br.readLine().trim();
        String mobile = br.readLine().trim();
        String dept = br.readLine().trim();
        String doctor = br.readLine().trim();

        if (patientMap.containsKey(id)) {
            out.append("Duplicate Patient ID.\n");
            return;
        }
        if (!name.matches("[a-zA-Z ]+")) {
            out.append("Invalid Patient Name.\n");
            return;
        }
        if (!mobile.matches("\\d{10}")) {
            out.append("Invalid Mobile Number.\n");
            return;
        }

        dept = dept.toUpperCase().trim();
        name = properCase(name);

        String[] data = {id, name, mobile, dept, doctor};
        idList.add(id);
        patientMap.put(id, data);
        out.append("Patient Registered Successfully.\n");
    }

    static void searchPatient(BufferedReader br) throws IOException {
        String sid = br.readLine().trim();
        if (patientMap.containsKey(sid)) {
            String[] p = patientMap.get(sid);
            out.append("Patient Details\n");
            out.append("ID : ").append(p[0]).append("\n");
        }
    }
}
```

Report

```

        out.append("Name : ").append(p[1]).append("\n");
        out.append("Mobile : ").append(p[2]).append("\n");
        out.append("Department : ").append(p[3]).append("\n");
        out.append("Doctor : ").append(p[4]).append("\n");
    } else {
        out.append("Patient Not Found\n");
    }
}

static void displayAll() {
    out.append("-----\n");
    out.append("ID Name Department\n");
    out.append("-----\n");
    for (String key : idList) {
        String[] p = patientMap.get(key);
        out.append(p[0]).append(" ").append(p[1]).append(" ").append(p[3]).append("\n");
    }
}

static void deptCount() {
    LinkedHashMap<String, Integer> count = new LinkedHashMap<>();
    for (String key : idList) {
        String[] p = patientMap.get(key);
        count.put(p[3], count.getOrDefault(p[3], 0) + 1);
    }
    out.append("Department Count\n");
    for (String d : count.keySet()) {
        out.append(d).append(" : ").append(count.get(d)).append("\n");
    }
}

static String properCase(String name) {
    String[] words = name.trim().split("\\s+");
    StringBuilder sb = new StringBuilder();
    for (String w : words) {
        String lower = w.toLowerCase();
        sb.append(Character.toUpperCase(lower.charAt(0))).append(lower.substring(1)).append(" ");
    }
    return sb.toString().trim();
}
}

```

INPUT

```

1
P101
Rahul Kumar
9876543210
Cardiology
Dr.Arun
1
P102
Priya
9876543211
Ortho
Dr.Meena
3
4
2
P101
5

```

OUTPUT

```

Patient Registered Successfully.
Patient Registered Successfully.
-----
ID Name Department
-----
P101 Rahul Kumar CARDIOLOGY
P102 Priya ORTHO
Department Count
CARDIOLOGY : 1
ORTHOD : 1
Patient Details

```

ID : P101
Name : Rahul Kumar
Mobile : 9876543210
Department : CARDIOLOGY
Doctor : Dr.Arun

Q2. Online Library Book Management System

CODE

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;
import java.util.Scanner;

public class LibraryManagementSystem {

    public static void main(String args[]) {

        ArrayList<String> bookId = new ArrayList<String>();
        ArrayList<String> title = new ArrayList<String>();
        ArrayList<String> author = new ArrayList<String>();
        ArrayList<String> category = new ArrayList<String>();
        ArrayList<String> publisher = new ArrayList<String>();

        HashMap<String,Integer> search = new HashMap<String,Integer>();
        HashSet<String> authors = new HashSet<String>();
        HashMap<String,Integer> count = new HashMap<String,Integer>();

        Scanner sc = new Scanner(System.in);

        int choice=0;

        while(choice!=7){

            System.out.println("1.Add Book");
            System.out.println("2.Search Book");
            System.out.println("3.Display Books");
            System.out.println("4.Display Unique Authors");
            System.out.println("5.Category Count");
            System.out.println("6.String Operations");
            System.out.println("7.Exit");

            choice=sc.nextInt();
            sc.nextLine();

            if(choice==1){

                System.out.print("Book ID : ");
                String id=sc.nextLine().trim();

                if(search.containsKey(id)){
                    System.out.println("Duplicate Book ID not allowed.");
                    continue;
                }

                System.out.print("Book Title : ");
                String t=sc.nextLine().trim();

                String arr[]=t.split(" ");
                String ans="";

                for(int i=0;i<arr.length;i++){

                    if(arr[i].length()>0){

                        String s=arr[i].toLowerCase();

                        ans=ans+
                            s.substring(0,1).toUpperCase()+
                            s.substring(1);
```

```
        if(i!=arr.length-1){
            ans=ans+" ";
        }

    }

}

System.out.print("Author : ");
String a=sc.nextLine().trim();

System.out.print("Category : ");
String c=sc.nextLine().trim().toUpperCase();

System.out.print("Publisher : ");
String p=sc.nextLine().trim();

if(p.equals("")){
    System.out.println("Publisher should not be empty.");
    continue;
}

bookId.add(id);
title.add(ans);
author.add(a);
category.add(c);
publisher.add(p);

search.put(id,bookId.size()-1);

authors.add(a);

if(count.containsKey(c)){
    count.put(c,count.get(c)+1);
}
else{
    count.put(c,1);
}

System.out.println("Book Added Successfully.");
}
else if(choice==2){

    System.out.print("Enter Book ID : ");
    String id=sc.nextLine().trim();

    if(search.containsKey(id)){

        int index=search.get(id);

        System.out.println("Book ID : "+bookId.get(index));
        System.out.println("Title : "+title.get(index));
        System.out.println("Author : "+author.get(index));
        System.out.println("Category : "+category.get(index));
        System.out.println("Publisher : "+publisher.get(index));

    }
    else{

        System.out.println("Book Not Available");

    }

}
```

```
}

else if(choice==3){

    if(bookId.size()==0){

        System.out.println("No Books Available.");

    }
    else{

        System.out.println("-----");
        System.out.printf("%-10s%-25s%-20s%-15s\n","ID","TITLE","AUTHOR","CATEGORY");
        System.out.println("-----");

        for(int i=0;i<bookId.size();i++){

            System.out.printf("%-10s%-25s%-20s%-15s\n",
                bookId.get(i),
                title.get(i),
                author.get(i),
                category.get(i));

        }

    }

}

else if(choice==4){

    if(authors.size()==0){

        System.out.println("No Authors Available.");

    }
    else{

        System.out.println("Unique Authors");

        Object list[]=authors.toArray();

        for(int i=0;i<list.length;i++){

            System.out.println(list[i]);

        }

    }

}

else if(choice==5){

    if(count.size()==0){

        System.out.println("No Categories Available.");

    }
    else{

        System.out.println("Category Wise Count");

        Object key[]=count.keySet().toArray();

    }

}
```

```
        for(int i=0;i<key.length;i++){

            System.out.println(key[i]+" : "+count.get(key[i]));

        }

    }

}

else if(choice==6){

    if(bookId.size()==0){

        System.out.println("No Books Available.");

    }

    else{

        for(int i=0;i<bookId.size();i++){

            String t=title.get(i);

            int words=0;

            if(t.length()>0){

                words=1;

                for(int j=0;j<t.length();j++){

                    if(t.charAt(j)==' '){

                        words++;

                    }

                }

            }

        }

        System.out.println("-----");
        System.out.println("Book ID : "+bookId.get(i));
        System.out.println("Title : "+t);
        System.out.println("Length : "+t.length());
        System.out.println("Words : "+words);
        System.out.println("Category : "+category.get(i));

    }

}

else if(choice==7){

    System.out.println("Thank You");

}

else{

    System.out.println("Invalid Choice");

}
```

Report

```
    }  
  
    System.out.println();  
  
    }  
  
    sc.close();  
  
    }  
  
}
```

INPUT

```
1  
8101  
java programming  
James Gosling  
Programming  
Oracle  
1  
8102  
data structures  
Mark Allen  
Computer Science  
Pearson  
2  
8101  
3  
4  
5  
6  
7
```

OUTPUT

(no output)